AskTheMenu: Playwright Testing Project

30 October 2026

1 Introduction

This project applies Playwright with Python to test AskTheMenu, a restaurant website. Four main features were selected: chat, menu browsing, the kitchen dashboard and QR table links. Each feature has three end-to-end tests: two positive cases and one negative case. The project also includes four API tests covering CRUD and two network-mocking tests.

Tools used: Python, Playwright, pytest and pytest-html. Tests were run on Chromium, Firefox and WebKit against the local website.

2 Features tested and left out

Feature	Scope	Reason
Chat assistant	Tested	Main way diners ask questions about the menu.
Menu browsing	Tested	Lets diners view dishes, prices and categories.
Kitchen dashboard	Tested	Lets staff view orders and change their status.
QR table links	Tested	Takes diners to the chat for the correct table.
AI answer accuracy and full ordering conversations	Left out	Replies vary, so the chat test checks that a reply arrives and is saved.
File/image uploads	Left out	Needs separate upload/storage testing; the four main features were prioritised.
Theme and appearance controls	Left out	Lower priority than the main user actions for this assignment.
Scanning QR codes with a phone camera	Left out	Tests use desktop browsers to check QR images and table links.

3 Test-case table

Mixed API cases include both successful requests and checks for incorrect input or access.

ID / feature	Type	What the test checks	Result
TC-01 / Chat	Positive	Open the chat page and check the input, send button and model choices.	Pass
TC-02 / Chat	Positive	Send a message, receive a reply and check that the conversation remains after refreshing.	Pass
TC-03 / Chat	Negative	Try empty and whitespace-only messages; neither should be sent.	Pass
TC-04 / Menu	Positive	Open the menu and check its heading, dishes, prices and category links.	Pass
TC-05 / Menu	Positive	Click each category and check that its section comes into view.	Pass
TC-06 / Menu	Negative	Open a missing menu page; expect HTTP 404 and a not-found message.	Pass
TC-07 / Kitchen	Positive	Open the kitchen dashboard and check its heading and filter buttons.	Pass
TC-08 / Kitchen	Positive	Check an order's details and move it through accepted, preparing and served.	Pass
TC-09 / Kitchen	Negative	Simulate a failed refresh/update; check the error message, unchanged order and Retry button.	Pass
TC-10 / QR routing	Positive	Check that all five tables have working QR images and chat links.	Pass
TC-11 / QR routing	Positive	Click Table 1 and check that its chat page opens.	Pass
TC-12 / QR routing	Negative	Open an invalid table link; check the error message and return to the menu.	Pass
TC-13 / API Create	Mixed	POST an order: expect 201 and correct order data. Empty items should return 400.	Pass
TC-14 / API Read	Mixed	GET the order list: expect 200 and the created order. A missing API route should return 404.	Pass
TC-15 / API Update	Mixed	PATCH an order status: expect 200 and the saved change. Invalid status should return 400.	Pass
TC-16 / API Delete	Mixed	DELETE an owned chat: expect 200 and confirm removal. A caller without a session should get 401.	Pass
TC-17 / Empty mock	Negative	Mock an empty chat response; check the notification and that another message can be sent.	Pass
TC-18 / Error mock	Negative	Mock HTTP 500; check the error notification and that another message can be sent.	Pass

4 API testing and network mocking

The API tests use Playwright's request context to send POST, GET, PATCH and DELETE requests. Create, read and update use orders; delete uses an owned chat. The tests check the response status and returned data, and confirm that changes were saved. They also check invalid input, a missing route and a request without a session.

The two network-mocking tests use page.route() to simulate an empty chat response and an HTTP 500 server error. Each test checks that a message appears, the input remains usable and the user can send another message.

5 Test results

There are 18 test cases. The 12 end-to-end tests and two mocking tests run in each of the three browsers, giving 42 browser runs. The four API tests run once, giving 46 runs in total. All 46 passed.

Browser / test group	Passed	Failed
Chromium	14	0
Firefox	14	0
WebKit	14	0
API tests (run once)	4	0
Total	46	0

6 Intentional bug demonstration

To show that a test can catch a website bug, the menu heading was deliberately changed from Restaurant Menu to Unavailable menu. TC-04 still expected Restaurant Menu, so it failed in Chromium, Firefox and WebKit. After the correct heading was restored, the same test passed in all three browsers. The script demonstrate_bug.py repeats this demonstration.

Bug	What was changed	Test result
Intentional bug	Menu heading changed from Restaurant Menu to Unavailable menu.	TC-04 failed in all three browsers. After restoring the heading, it passed in all three.

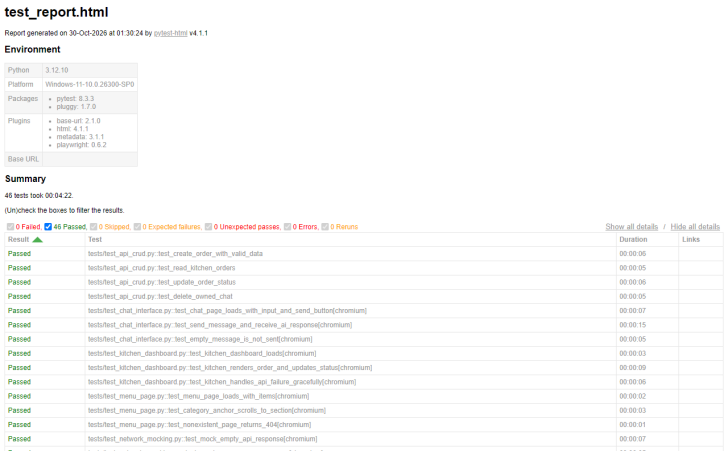
7 Team contributions

Member	Work
Member 1: [Name / Student ID]	Chat and menu tests; network mocking tests; bug demonstration and report.
Member 2: [Name / Student ID]	Kitchen and QR tests; API tests; screenshots and test reports.

Both members should be able to explain every test during the final demo. DEMO_GUIDE.md provides a walkthrough for practice.

8 Submitted files

The submission ZIP contains the Python test files, an HTML report with all test results, screenshots, the intentional-bug results and this report. The main results are in reports/test_report.html. The bug demonstration is recorded in reports/bug_injected.html and reports/bug_fixed.html. Setup and run instructions are in README.md.



test_report.html

Report generated on 30-Oct-2026 at 01:30:24 by pytest-html v4.1.1

Environment

Python	3.12.10
Platform	Windows-11-10.0.26300-SP0
Packages	• pytest: 8.3.3 • pluggy: 1.7.0
Plugins	• base-url: 2.1.0 • html: 4.1.1 • metadata: 3.1.1 • playwright: 0.6.2
Base URL	

Summary

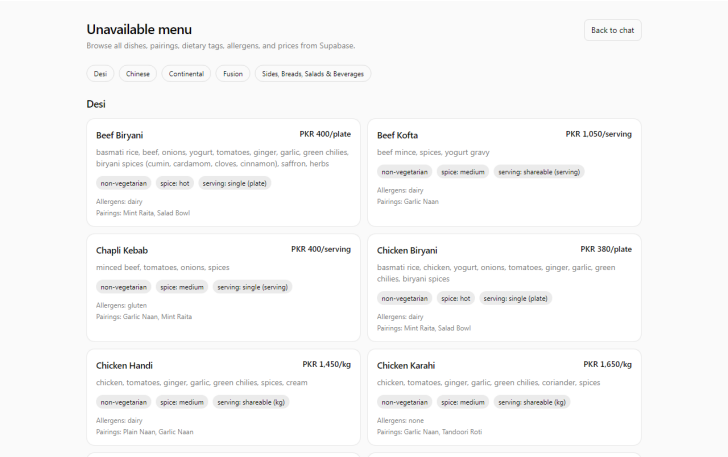
46 tests took 00:04:22.

(Un)check the boxes to filter the results.

0 Failed, 46 Passed, 0 Skipped, 0 Expected failures, 0 Unexpected passes, 0 Errors, 0 Reruns — Show all details / Hide all details

Result	Test	Duration	Links
Passed	tests/test_api_crud.py::test_create_order_with_valid_data	00:00:06	
Passed	tests/test_api_crud.py::test_read_kitchen_orders	00:00:05	
Passed	tests/test_api_crud.py::test_update_order_status	00:00:06	
Passed	tests/test_api_crud.py::test_delete_owned_chat	00:00:05	
Passed	tests/test_chat_interface.py::test_chat_page_loads_with_input_and_send_button[chromium]	00:00:07	
Passed	tests/test_chat_interface.py::test_send_message_and_receive_ai_response[chromium]	00:00:15	
Passed	tests/test_chat_interface.py::test_empty_message_is_not_sent[chromium]	00:00:05	
Passed	tests/test_kitchen_dashboard.py::test_kitchen_dashboard_loads[chromium]	00:00:03	
Passed	tests/test_kitchen_dashboard.py::test_kitchen_renders_order_and_updates_status[chromium]	00:00:09	
Passed	tests/test_kitchen_dashboard.py::test_kitchen_handles_api_failure_gracefully[chromium]	00:00:06	
Passed	tests/test_menu_page.py::test_menu_page_loads_with_items[chromium]	00:00:02	
Passed	tests/test_menu_page.py::test_category_anchor_scrolls_to_section[chromium]	00:00:03	
Passed	tests/test_menu_page.py::test_nonexistent_page_returns_404[chromium]	00:00:01	
Passed	tests/test_network_mocking.py::test_mock_empty_api_response[chromium]	00:00:07	

Figure 1: HTML test report showing the passing results.



Unavailable menu

Browse all dishes, pairings, dietary tags, allergens, and prices from Supabase.

Back to chat

Desi | Chinese | Continental | Fusion | Sides, Breads, Salads & Beverages

Desi

Beef Biryani — PKR 400/plate
basmati rice, beef, onions, yogurt, tomatoes, ginger, garlic, green chilies, biryani spices (cumin, cardamom, cloves, cinnamon), saffron, herbs
non-vegetarian | spice: hot | serving: single (plate)
Allergens: dairy
Pairings: Mint Raita, Salad Bowl

Beef Kofta — PKR 1,050/serving
beef mince, spices, yogurt gravy
non-vegetarian | spice: medium | serving: shareable (serving)
Allergens: dairy
Pairings: Garlic Naan

Chapli Kebab — PKR 400/serving
minced beef, tomatoes, onions, spices
non-vegetarian | spice: medium | serving: single (serving)
Allergens: gluten
Pairings: Garlic Naan, Mint Raita

Chicken Biryani — PKR 380/plate
basmati rice, chicken, yogurt, onions, tomatoes, ginger, garlic, green chilies, biryani spices
non-vegetarian | spice: hot | serving: single (plate)
Allergens: dairy
Pairings: Mint Raita, Salad Bowl

Chicken Handi — PKR 1,450/kg
chicken, tomatoes, ginger, garlic, green chilies, spices, cream
non-vegetarian | spice: medium | serving: shareable (kg)
Allergens: dairy
Pairings: Plain Naan, Garlic Naan

Chicken Karahi — PKR 1,650/kg
chicken, tomatoes, ginger, garlic, green chilies, coriander, spices
non-vegetarian | spice: medium | serving: shareable (kg)
Allergens: none
Pairings: Garlic Naan, Tandoori Roti

Figure 2: The deliberately incorrect menu heading used to demonstrate a failing test.